Math 150-A Homework 1-9

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Homework 1

1. Consider the following questions concerning bounded set.

1. State the definition of a bounded set $S \subset \mathbb{R}$.

   If S is bounded in $\mathbb{R}$, then there exists a real number $M < \infty$ such that for all $x \in S$ we have
   $$|x| \leq M$$

2. Show that $S = \{1, ..., 10\} \cup (-1, 1)$ is bounded.

   By the definition, choose $M = 10$. We have that for all $x \in \{1, .., 10\} \cup (-1, 1)$, we get $|x| \leq M$.

3. Show that the set of natural numbers $\mathbb{N}$ is an unbounded subset of $\mathbb{R}$

   Assume for contradiction that $\mathbb{N}$ is bounded, then for all $x \in \mathbb{N}$ we have a number $M < \infty$ such that $|x| \leq M$ for all x. Choose $x_0 := \lceil M \rceil + 1$ we get that $x_0 \in \mathbb{N}$ but $|x_0| > M$. A contradiction.

4. Show that the union of two bounded sets is bounded.

   Suppose we have two bounded sets A and B. Thus we have real numbers $M_A, M_B < \infty$ such that
   $$x \in A \implies |x| < M_A, \quad \text{and} \quad x \in B \implies |x| < M_B$$
   Define the union as $C := A \cup B$ and define $M_C = \max\{M_A, M_B\}$. For $x \in C$ we get that
   $$x \in A \implies |x| < M_A \leq M_C, \quad \text{and} \quad x \in B \implies |x| < M_B \leq M_C$$
   Meaning that the union of two bounded sets is bounded.

2. We have triangle inequality: $|x + y| \leq |x| + |y|$ for all $x, y \in \mathbb{R}$.

1. When the inequality becomes equality, i.e. $|x + y| = |x| + |y|$?

   Let $a = |x + y|$ and $b = |x| + |y|$. Computing the squares we get
   $$a^2 = (x + y)^2 = x^2 + 2xy + y^2 \quad b^2 = x^2 + 2|xy| + y^2$$
   If equality holds, we have $a^2 - b^2 = 0$ and thus
   $$x^2 + 2xy + y^2 - (x^2 + 2|xy| + y^2) = 0 \iff 2xy - 2|xy| = 0 \iff xy = |xy|$$
   By definition of the absolute value we get that $xy \geq 0$ or
   $$|x + y| = |x| + |y| \iff xy \geq 0$$

2. When it becomes strict inequality, i.e. $|x+y| < |x| + |y|$?

   Continuing from part 1, but assume that $b > a$ thus $b^2 - a^2 > 0$. Following similar algebra from problem 1, we get.

   $$|xy| > xy$$

   Thus

   $$|x+y| < |x| + |y| \iff xy < 0$$

3. Show that $|x-y| \leq |x| + |y|$ for all $x, y \in \mathbb{R}$ via the triangle inequality

   Since $x - y = x + (-y)$ we get the following

   $$|x-y| = |x + (-y)| \leq |x| + |-y| = |x| + |y|$$

   $$|x-y| \leq |x| + |y|$$

4. Show that $||x| - |y|| \leq |x-y|$ for all $x, y \in \mathbb{R}$ via the triangle inequality

   Start with the triangle inequality

   $$|x| = |x + y - y| \leq |x-y| + |y|$$

   Thus

   $$|x| - |y| \leq |x-y|$$

   Since the inequality must hold symmetrically for x and y, take the absolute values of both sides

   $$||x| - |y|| \leq |x-y|$$

3. Consider the following questions concerning the definition of convergent sequences

1. State the definition of convergent sequences

   For $a_n : \mathbb{N} \to \mathbb{R}$, then $\{a_n\}$ is a sequence that converges to a iff for all $\varepsilon > 0$, there exists N such that for all $n \geq N$ we have $|a_n - a| < \varepsilon$

2. Use the definition to show that if $\lim_{n\to\infty} a_n = L$ then $\lim_{n\to\infty} |a_n| = |L|$

   Fix $\varepsilon > 0$. Suppose $\{a_n\}_{n\to\infty} = L$. Then there exists N such that for all $n \geq N$ then $|a_n - L| < \varepsilon$. Choose $N' = N$ and recall from 2.4 that

   $$||x| - |y|| \leq |x-y|$$

   Thus we have N' such that for all $n \geq N$

   $$||a_n| - |L|| \leq |a_n - L| < \varepsilon$$

   Since ε was arbitrary the sequence converges to L.

3. Is it true that that if $\lim_{n\to\infty} |a_n| = |L|$ then $\lim_{n\to\infty} a_n = L$? Why?

   No, it would suffice to provide a counter example. Suppose $a_n = (-1)^n$. Then $|a_n| = 1$ for all n and thus.

   $$\lim_{n\to\infty} |a_n| = 1$$

   But $\lim_{n\to\infty} a_n$ diverges.

4. Try to mimic the class examples to show convergence of the following sequences by definition.

1. $\lim_{n\to\infty} \frac{1}{n^2} = 0$

   Fix $\varepsilon > 0$. Choose $N = \left\lceil \frac{1}{\sqrt{\varepsilon}} \right\rceil$. Thus we get that for all $n > N$ we get

   $$|a_n - 0| = \left|\frac{1}{n^2}\right| \leq \left|\frac{1}{1/\sqrt{\varepsilon}^2}\right| = \varepsilon$$

   Thus by definition $\lim_{n\to\infty} \frac{1}{n^2} = 0$

2. $\lim_{n\to\infty} \frac{1}{2^n} = 0$

   Fix $\varepsilon > 0$. Choose $N = \left\lceil \log_2\left(\frac{1}{\varepsilon}\right) \right\rceil$. Thus we get that for all $n > N$ we get

   $$|a_n - 0| = \left|\frac{1}{2^n}\right| \leq \left|\frac{1}{2^{\log_2\left(\frac{1}{\varepsilon}\right)}}\right| = \varepsilon$$

3. Let $0 < a < 1$. Show that $\lim_{n\to\infty} n^2 a^n = 0$

   Since $a < 1$, we get $\frac{1}{a} > 1$. Thus there exists $c \in \mathbb{R}$ such that $1 < c < \frac{1}{a}$ and thus $0 < ca < 1$. Next, claim there exists $M_1 < \infty$, $n^2 \leq c^n$ for all $n \geq M_1$. Employing a ratio test

   $$\frac{c^{n+1}}{(n+1)^2} \cdot \frac{n^2}{c^n} = c\left(\frac{n}{n+1}\right) \to c > 1$$

   Thus the claim is true and for all $n \geq M_1$ we have

   $$n^2 a^n \leq c^n a^n = (ac)^n$$

   Thus if we show the limit of ac^n is 0, then by the squeeze theorem $n^2 a^n$ is also 0. For some fixed ε we want $|(ac)^n| < \varepsilon$. Recall that since $ac < 1$ then $\ln ac < 0$

   $$\iff n \ln(ac) < \ln \varepsilon$$

The inequality switches when dividing by $\ln ac$ and thus

$$n > \frac{\ln \varepsilon}{\ln ac}$$

So choose an integer

$$M_2 > \frac{\ln \varepsilon}{\ln ac}$$

Thus for all $n \geq M_2$ we get $ac^n < \varepsilon$. We then choose the constant where both the limit and $n^2 \leq c^n$ are satisfied.

$$N := \max\{M_1, M_2\}$$

Finally, we get that for all $\varepsilon > 0$ we have an integer N such that for all $n \geq N$ we get $|a^n n^2 - 0| < \varepsilon$.

Homework 2

Try to mimic the class examples to show convergence of the following sequences by definition.

(Problems 4.1,4.2, and 4.3) were taken from Homework 1)
(4) Try this one if you are ambitious (it won't be on the quiz): Let $0 < a < 1$ and $k \in \mathbb{Z}_+$. Show that $\lim_{n\to\infty} n^k a^n$

(Very similar to problem 3 but for $k \in \mathbb{Z}_+$ instead of $k = 2$ but the logic still holds.) Since $a < 1$ then $\frac{1}{a} > 1$ and there exists c such that $1 < c < \frac{1}{a}$ and thus $0 < ca < 1$. Next claim that for some large M_1 we have $c^n \geq n^k$ for all $n \geq M$. Employing the ratio test

$$\frac{c^{n+1}}{(n+1)^k} \cdot \frac{n^k}{c^n} = c\left(\frac{n}{n+1}\right)^k \to c > 1$$

Thus the claim is true and for all $n \geq M_1$ we have

$$n^k a^n \leq c^n a^n = (ac)^n$$

From here the proof is exactly the same as part 4.3 from Homework 1. Thus for all $\varepsilon > 0$ we have an integer N such that for all $n \geq N$ we get $|a^n n^k - 0| < \varepsilon$.

2. Show that if $\{a_n\}$ is bounded and $\{b_n\}$ converges to 0, then $\lim_{n\to\infty}(a_n b_n) = 0$.

Since $\{a_n\}$ is bounded, then for all $n \in \mathbb{N}$ we get that there exists $B < \infty$ such that $|a_n| \leq B$. Since $\{b_n\}$ converges to 0 then for every $\varepsilon > 0$ there exists $N < \infty$ such that for all $n \geq N$ we have $|b_n| < \dfrac{\varepsilon}{B}$. Thus we get that $|B\, b_n| \geq |a_n b_n|$ for all $n \in \mathbb{N}$. Next we want to evaluate the following limit $\lim_{n\to\infty} B\, b_n$ to be 0 and then use squeeze theorem to finalize the proof. Fix $\varepsilon > 0$. We already have N such that for all $n \geq N$ we get $|Bb_n - 0| < \varepsilon$. Thus $\lim_{n\to\infty} B\, b_n = 0$ By squeeze theorem

$$\lim_{n\to\infty} |B\, b_n| \geq \lim_{n\to\infty} |a_n b_n| \geq 0$$
$$\implies 0 \geq \lim_{n\to\infty} a_n b_n \geq 0$$

3. Consider the following questions.

1. Let $\{a_n\}$ and $\{b_n\}$ be two convergent sequences. Let α and $\beta \in \mathbb{R}$ be two constants. Apply the product and sum laws to show that.

$$\lim_{n\to\infty}(\alpha a_n + \beta b_n) = \alpha \lim_{n\to\infty} a_n + \beta \lim_{n\to\infty} b_n$$

Since α and β are constants, we get that by the product law

$$\lim_{n\to\infty} \alpha a_n = \alpha \lim_{n\to\infty} a_n$$

$$\lim_{n\to\infty} \beta b_n = \beta \lim_{n\to\infty} b_n$$

The sum law states that if a_n and b_n are convergent series then

$$\lim_{n\to\infty} (\alpha a_n + \beta b_n) = \lim_{n\to\infty} \alpha a_n + \lim_{n\to\infty} \beta b_n$$

Thus we can derive

$$\lim_{n\to\infty} \alpha a_n + \lim_{n\to\infty} \beta b_n = \alpha \lim_{n\to\infty} a_n + \beta \lim_{n\to\infty} b_n$$

2. Use part (4) from problem 1 and part (1) of problem 3 (or rather its remark) to show that

$$\lim_{n\to\infty} (p(n)a^n) = 0$$

Where $p(n) = \sum_{k=0}^{m} a_k n^k$ is a polynomial in degree n and $0 < a < 1$ is a constant. Rewriting the limit we get

$$\lim_{n\to\infty} \sum_{k=0}^{m} a^n a_k n^k$$

Using linearity of the sum and the limit we can say that the limit of the finite sum is the same as the finite sum of limits and thus

$$\sum_{k=0}^{m} a_k \lim_{n\to\infty} a^n n^k$$

By part (4) of problem 1 we get that for all $k \in \mathbb{Z}_+$ we get $\lim_{n\to\infty} a^n n^k = 0$. Thus

$$\sum_{k=0}^{m} a_k \cdot 0 = \sum_{k=0}^{m} 0 = 0$$

4. Consider the following questions:

1. Find an example where $\{a_n\}$ and $\{b_n\}$ are both divergent but $\{a_n + b_n\}$ is convergent.

   Let $\{a_n\}$ be any divergent sequence where $a_n \neq \infty$ for any $n \in \mathbb{N}$. Then choose $b_n = -a_n$ for all n. Thus $\{b_n\}$ must also be a divergent sequence. Then we get that $a_n - b_n = 0$ for all n. Thus the constant sequence $\{0\} \to 0$ is convergent.

2. If $\{a_n\}$ convergent and $\{b_n\}$ is divergent, can $\{a_n + b_n\}$ be convergent?

   Assume that $\{a_n + b_n\}$ is convergent. Thus by the product rule, $\{-a_n\}$ is also convergent. By the sum rule

$$\lim_{n\to\infty}(a_n + b_n) + \lim_{n\to\infty}(-a_n) = \lim_{n\to\infty}(a_n + b_n - a_n)$$

Homework 3

1. Try to write down the proof of the sum and product limit laws without consulting any resources

Sum law

Suppose we have two convergent sequences and their following limits

$$\lim_{n\to\infty} a_n = L_a \quad \text{and} \quad \lim_{n\to\infty} b_n = L_b$$

Thus for each of the limits we have for $\varepsilon > 0$ there are constants $M_a, M_b < \infty$ such that for all $n \geq M_a$ and $n \geq M_b$ respectively, we have

$$|L_a - a_n| < \frac{\varepsilon}{2} \quad |L_b - b_n| < \frac{\varepsilon}{2}$$

Choose $N := \max\{M_a, M_b\}$ such that the initial, individual limits hold. Next, claim this result of the following limit

$$\lim_{n\to\infty} (a_n + b_n) = L_a + L_b$$

This is true if and only if that for some N we have for all $n \geq N$ we have

$$|(a_n + b_n) - (L_a + L_b)| < \varepsilon$$

Rewrite the absolute value and using triangle inequality

$$|a_n - L_a + b_n - L_b| \leq |a_n - L_a| + |b_n - L_b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Thus the claim holds and we get

$$\lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n = L_a + L_b = \lim_{n\to\infty} (a_n + b_n)$$

Product law

Fix $\varepsilon > 0$. Under the same assumptions as for the sum rule, there also exists M_a and M_b such that

$$|L_a - a_n| < \frac{\varepsilon}{2(L_a + 1)} \qquad |L_b - b_n| < \frac{\varepsilon}{2K}$$

Where K is the constant that bounds the sequence $\{b_n\}$ such that $|b_n| \leq K$ for all n. We know K exists because every convergent sequence is bounded. Choose $N := \max\{M_a, M_b\}$ such that the initial, individual limits hold. Next, claim this result of the following limit

$$\lim_{n\to\infty} (a_n \cdot b_n) = L_a \cdot L_b$$

This is true if and only if that for some N we have for all $n \geq N$ we have

$$|a_n \cdot b_n - L_a \cdot L_b| < \varepsilon$$

Rewrite the absolute value

$$|a_n b_n - b_n L_a + b_n L_a - L_a L_b| = |b_n(a_n - L_a) + L_a(b_n - L_b)|$$

By triangle inequality

$$\leq |b_n(a_n - L_a)| + |L_a(b_n - L_b)|$$
$$= K\frac{\varepsilon}{2K} + L_a\frac{\varepsilon}{2(L_a+1)} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Thus the claim holds and we get

$$\lim_{n\to\infty} a_n \cdot \lim_{n\to\infty} b_n = L_a \cdot L_b = \lim_{n\to\infty} (a_n \cdot b_n)$$

2. Consider the following sequences. If convergent, compute their limits and justify your steps. If divergent, explain your reasoning. You may directly use the class examples or the examples from previous homework assignments.

1. $\left\{\dfrac{n3^n + n^2 2^n + 5n}{5^n + n^3}\right\}$

   Multiply the numerator and denominator by $\left(\frac{1}{5}\right)^n$ and we get

   $$\frac{n3^n + n^2 2^n + 5n}{5^n + n^3}\frac{\left(\frac{1}{5}\right)^n}{\left(\frac{1}{5}\right)^n} = \frac{n\left(\frac{3}{5}\right)^n + n^2\left(\frac{2}{5}\right)^n + 5n\left(\frac{1}{5}\right)^n}{\left(\frac{5}{5}\right)^n + n^3\left(\frac{3}{5}\right)}$$

   Recall from problem 1 on homework 2 that for $0 < a < 1$ and $k \in \mathbb{Z}_+$ we get $\lim\limits_{n\to\infty} a^n n^k = 0$. Applying to the sequence we get

   $$\lim_{n\to\infty} \frac{n\left(\frac{3}{5}\right)^n + n^2\left(\frac{2}{5}\right)^n + 5n\left(\frac{1}{5}\right)^n}{\left(\frac{5}{5}\right)^n + n^3\left(\frac{3}{5}\right)} = \frac{0+0+5(0)}{1+0} = 0$$

   So the sequence converges to 0

2. $\left\{\sqrt{n+2} - \sqrt{n}\right\}$

   Multiply the numerator and denominator by the conjugate: $\sqrt{n+2} + \sqrt{n}$

   $$\left(\sqrt{n+2} - \sqrt{n}\right) \cdot \frac{\sqrt{n+2} + \sqrt{n}}{\sqrt{n+2} + \sqrt{n}}$$
   $$= \frac{n+2-n}{\sqrt{n+2} + \sqrt{n}} = \frac{2}{\sqrt{n+2} + \sqrt{n}}$$

   Since $\sqrt{n+2} + \sqrt{n} > \sqrt{n}$, then we get

   $$\frac{2}{\sqrt{n+2} + \sqrt{n}} < \frac{2}{\sqrt{n}}$$

It would suffice to show that $\frac{1}{\sqrt{n}}$ converges to 0. Fix, $\varepsilon > 0$ Choose $N = \frac{1}{\varepsilon^2}$. Thus we get for all $n > N$ we get

$$n > \frac{1}{\varepsilon^2}$$

$$\left| \frac{1}{\sqrt{n}} - 0 \right| < \varepsilon$$

3. $\left\{ \frac{n^2+1}{n} \right\}$

Consider the inequality

$$\frac{n^2+1}{n} \geq \frac{n^2}{n} = n$$

Since n is unbounded, then the sequence is also unbounded.

3. Use the example $e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n$ and the formula

$$1 + \frac{3}{n} = \left(1 + \frac{1}{n}\right)\left(1 + \frac{1}{n+1}\right)\left(1 + \frac{1}{n+2}\right)$$

to show that

$$\lim_{n\to\infty} \left(1 + \frac{3}{n}\right)^n = e^3$$

Performing a substitution of the formula into the limit, problem basically asks us to evaluate

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n \left(1 + \frac{1}{n+1}\right)^n \left(1 + \frac{1}{n+2}\right)^n$$

Recall the product rule, we get that the limit can be rewritten as

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n \cdot \lim_{n\to\infty} \left(1 + \frac{1}{n+1}\right)^n \cdot \lim_{n\to\infty} \left(1 + \frac{1}{n+2}\right)^n$$

The first limit, by definition, gets us

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right) = e$$

Rewriting the second limit and re-indexing $n + 1 \mapsto m$ we get

$$= \lim_{n\to\infty} \frac{\left(1 + \frac{1}{n+1}\right)^{n+1}}{\left(1 + \frac{1}{n+1}\right)}$$

$$= \frac{\lim_{m\to\infty}\left(1+\frac{1}{m}\right)^m}{\lim_{m\to\infty}\left(1+\frac{1}{m}\right)}$$
$$= \frac{e}{1} = e$$

Doing the same for the 3rd limit and re-indexing $n+2 \mapsto m$ we get

$$= \lim_{n\to\infty}\left(1+\frac{1}{n+2}\right)^n$$
$$= \lim_{n\to\infty}\frac{\left(1+\frac{1}{n+2}\right)^{n+2}}{\left(1+\frac{1}{n+2}\right)^2}$$
$$= \frac{\lim_{m\to\infty}\left(1+\frac{1}{m}\right)^m}{\lim_{m\to\infty}\left(1+\frac{1}{m}\right)^2}$$
$$= \frac{e}{1^2} = e$$

Thus our limit gets us

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n\left(1+\frac{1}{n+1}\right)^n\left(1+\frac{1}{n+2}\right)^n = e\cdot e\cdot e = e^3$$

4. Define a sequence by $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2+a_n}$ for all $n \geq 1$. Compute the limit.

For the base case, we see that $a_1 = \sqrt{2} \leq 2$. For the inductive step suppose that for some k, we have $a_{k+1} = \sqrt{2+a_k}$ where $a_k \leq 2$. Thus we get that

$$a_k + 2 \leq 4 \implies \sqrt{a_k+2} \leq 2 \implies a_{k+1} \leq 2$$

Thus the sequence is bounded above by 2. To show that the sequence is monotonically increasing it would suffice to show by induction that $\frac{a_{n+1}}{a_n} > 1$ for all n. For the base case

$$\frac{a_2}{a_1} = \frac{\sqrt{2+\sqrt{2}}}{\sqrt{2}} = \sqrt{\frac{2+\sqrt{2}}{2}} > \sqrt{\frac{2}{2}} = 1$$

For the inductive step, suppose that $\frac{a_k}{a_{k-1}} > 1$ is true for some k. Then we have

$$\frac{a_{k+1}}{a_k} = \frac{\sqrt{2+a_k}}{\sqrt{2+a_{k-1}}} = \sqrt{\frac{2+a_k}{2+a_{k-1}}} > \sqrt{\frac{2}{2}} = 1$$

Since it's bounded above and monotonically increasing, the limit exists. Let L be the limit defined by

$$\lim_{n\to\infty} a_{n+1}$$

Expanding the limit we get

$$\lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \sqrt{a_n + 2} = \sqrt{\lim_{n\to\infty} a_n + 2}$$

The limit must also be defined by $\lim_{n\to\infty} a_n$ so substituting we get

$$L = \sqrt{L + 2} \iff L = 2$$

Homework 4

1. Write down the definitions of limit point, limit set, closure, and closed set without consulting any resources.

1. If $x \in X$ is a limit point of some set $S \subset X$, Then for all $\varepsilon > 0$ there exists $x_n \in S \backslash \{x\}$ such that $d(x_n, x) < \varepsilon$
2. A limit set S' of S is the set of all limit points of S
3. The closure of S is the smallest closed set that contains S. It's the union of the set S and S'.
4. A closed set is a set that contains all of its limit points. The compliment of an open set.

2. Apply Theorem 1.14 (in my lecture notes) to find the closure of the set

$$S = \left\{ \frac{1}{n} : n \in \mathbb{Z}_+ \right\}$$

Remark: note here you need to find all the limit points of S.

Recall that the closure of S is just $S \cup S'$. And $S' = \{0\}$ because it's a limit point of S. We know this because $\lim_{n\to\infty} \frac{1}{n} = 0$. To show that 0 is the *only* limit point of S, assume for contradiction that there is some fixed number $\frac{1}{m} \neq 0$ that is a limit point of S. Thus for all $\varepsilon > 0$ there exists N such that for all $n \geq N$

$$\left| \frac{1}{m} - \frac{1}{n} \right| < \varepsilon$$

However, for $\varepsilon = \frac{1}{2|m|}$, we get that for all $n > 2m$ we get that

$$\left| \frac{1}{m} - \frac{1}{n} \right| > \frac{1}{2|m|}$$

Thus this fails for some ε and we get a contradiction. Thus $\{0\}$ is the only limit point of S.

Try the following questions: (i) show that a closed interval is a closed set; (ii) show that the set of integers $\mathbb{Z}$ is a closed set; (iii) based on your proof of part (i), find an unbounded set that has no limit point.

(i) Suppose that x is a limit point in the closed interval $[a, b]$, then we need to show that $x \in [a, b]$. Meaning that for any $\varepsilon > 0$ we have some x_n such that $|x_n - x| < \varepsilon$. Assume for contradiction that $x \notin [a, b]$. Then

Case 1: $x < a$
Choose $\varepsilon = \dfrac{a - x}{2} > 0$ we get that the interval
$(x - \varepsilon, x + \varepsilon) \cap [a, b] = \varnothing$ because

$$x - \frac{a - x}{2} < x < x + \frac{a - x}{2} < a$$

Case 2: $b < x$
Similarly, Choose $\varepsilon = \dfrac{x - b}{2} > 0$ we get that the interval
$(x - \varepsilon, x + \varepsilon) \cap [a, b] = \varnothing$ because

$$b < x - \frac{x - b}{2} < x < x + \frac{x - b}{2}$$

Thus we get a contradiction and $x \in [a, b]$. Since x was arbitrary, this means that every limit point is in a closed interval is contained in the interval, thus

(ii) (This would be so much easier if I was allowed to show that it's the compliment of an open set.) Suppose that $x \in \mathbb{R}$ is a limit point of $\mathbb{Z}$. Assume for contradiction that x exists and $x \notin \mathbb{Z}$. Then $\lfloor x \rfloor < x < \lceil x \rceil$. Choose $\varepsilon = \dfrac{x^*}{2}$ where $x^* := \min\{x - \lfloor x \rfloor, \lceil x \rceil - x\}$. We get that

$$(x - \varepsilon, x + \varepsilon) \cap \mathbb{Z} = \varnothing$$

Which contradicts the limit point definition. Thus $x \notin \mathbb{R} \backslash \mathbb{Z}$. Next assume that $x \in \mathbb{Z}$. Choose $\varepsilon < 1$. Again no other integer is in the range $(x - \varepsilon, x + \varepsilon)$. So $x \notin \mathbb{Z}$ then (in a technicality) $\mathbb{Z}$ contains all of it's limit points (which there are none). Meaning $\mathbb{Z}$ is closed.

(iii) From part (ii) we get that $\mathbb{Z}$ is an unbounded set with no limit points. Proof of boundlessness of $\mathbb{Z}$: Suppose $\mathbb{Z}$ is bounded. There exists $B < \infty$ such that for all $x \in \mathbb{Z}$ we get $|x| < B$. Choose $\lceil B \rceil + 1 \in \mathbb{Z}$. But $\lceil B \rceil + 1 \not< B$. A contradiction.

4. Try to mimic the proof of Theorem 1.15 to do the following problem: Let $\{I_n\}_{n \in \mathbb{Z}_+}$ be a sequence of closed intervals, and denote by λ_n the length of I_n. Assume that it's decreasing in the sense that

$$I_1 \supset I_2 \supset \cdots I_n \supset I_{n+1} \supset \cdots$$

Show that (i) $\lim \lambda_n$ exists; (ii) if $\lim \lambda_n > 0$, then the intersection of those intervals. $\bigcap_{n \in \mathbb{Z}_+} I_n$ is a closed interval with length $\lim \lambda_n$

Writing the interval as $[a_n, b_n] := I_n$, recall that $\lambda_n = b_n - a_n$, where $b > a$ and

$$a_1 \leq a_2 \leq \cdots \leq a_n \cdots \leq b_n \leq \cdots \leq b_2 \leq b_1$$

(i) Meaning that $\lambda_n \geq 0$ for all n. And since λ_n is bounded from below and decreasing we get that the limit exists.

(ii) Thus since $\lim \lambda_n > 0$ we get that $\lim a_n < \lim b_n$. Let $a := \lim a_n$, $b := \lim b_n$, and $\lambda := \lim \lambda_n$. So $b > a$ and $b - a = \lambda$. Finally we need to show that the intersection is a closed interval and it would suffice to show that the intersection any amount of closed intervals is always closed.
Let $\{I_n\}_{n \in \mathbb{Z}_+}$ be an arbitrary collection of closed intervals in $\mathbb{R}$ and let

$$I := \bigcap_{n \in \mathbb{Z}_+} I_n$$

Let x be a limit point of I. Since $I \subset I_n$ for each n and each I_n is a closed interval (it contains all of its limit points) then we have

$$x \in \bigcap_{n \in \mathbb{Z}_+} I_n = I$$

Thus I contains all of its limit points.

Homework 5

1. Let $f : I \to \mathbb{R}$ be a function defined on an interval $I \subset \mathbb{R}$. Write down the definitions of $\lim\limits_{x\to\xi} f(x) = A$, $\lim\limits_{x\to\xi_+} f(x) = A$, $\lim\limits_{x\to\xi_-} f(x) = A$, and the definition of f *is continuous at* ξ . What does it mean by saying f *is continuous on* I*?*

1. $\lim\limits_{x\to\xi} f(x) = A$
   For all $\varepsilon > 0$ there exists $\delta > 0$ such that
   $$0 < |x - \xi| < \delta \implies |f(x) - A| < \varepsilon$$

2. $\lim\limits_{x\to\xi_+} f(x) = A$
   For all $x > \xi$. For all $\varepsilon > 0$ there exists $\delta > 0$ such that
   $$0 < x - \xi < \delta \implies |f(x) - A| < \varepsilon$$

3. $\lim\limits_{x\to\xi_-} f(x) = A$
   For all $x < \xi$. For all $\varepsilon > 0$ there exists $\delta > 0$ such that
   $$0 < \xi - x < \delta \implies |f(x) - A| < \varepsilon$$

4. If f is continuous at ξ then
   $$\lim_{x\to\xi} f(x) = f(\xi)$$

5. If f is continuous on I then f is continuous for all $x \in I$

2. Show by definition that the function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = |x|$, is continuous on $\mathbb{R}$.

Fix $\varepsilon > 0$ and let $\xi \in \mathbb{R}$ be arbitrary. Choose $\delta = \varepsilon$. Then,

$$|f(x) - f(\xi)| = ||x| - |\xi||$$

Using reverse triangle inequality

$$||x| - |\xi|| \leq |x - \xi| < \delta = \varepsilon$$

Since ξ was arbitrary, $f(x)$ is continuous at every $\xi \in \mathbb{R}$, hence f is continuous on $\mathbb{R}$.

3. Define $f : \mathbb{R} \to \mathbb{R}$ to be

$$f(x) = \begin{cases} 1 & \textbf{if } x \geq 0 \\ -1 & \textbf{if } x < 0 \end{cases}$$

1. Show by definition that f is continuous at every $x \neq 0$

   Fix $\varepsilon > 0$. Choose $\delta = \min\{\varepsilon, |\frac{\xi}{2}|\}$. Suppose $|x - \xi| < \delta$. If $\xi < 0$ we force $x < 0$ because $(\xi - \delta, \xi + \delta) \cap [0, \infty) = \varnothing$. Thus

   $$|f(x) - f(\xi)| = |(-1) - (-1)| = 0 < \delta = \varepsilon$$

   Similarly, if $\xi > 0$ we force $x > 0$ because $(\xi - \delta, \xi + \delta) \cap (-\infty, 0] = \varnothing$. Thus

   $$|f(x) - f(\xi)| = |1 - 1| = 0 < \delta = \varepsilon$$

2. Compute by definition $\lim\limits_{x \to 0_+} f(x)$ and $\lim\limits_{x \to 0_-} f(x)$

   Claim that $\lim\limits_{x \to 0_-} f(x) = -1$ for all $x < 0$ and $\varepsilon > 0$. Choose $\delta = \varepsilon$. Thus

   $$0 < 0 - x < \delta \implies |-1 - (-1)| = 0 < \varepsilon$$

   Similarly, claim that $\lim\limits_{x \to 0_+} f(x) = 1$ for all $x > 0$ and $\varepsilon > 0$. Choose $\delta = \varepsilon$. Thus

   $$0 < x - 0 < \delta \implies |1 - 1| = 0 < \varepsilon$$

   Therefore $\lim\limits_{x \to 0_-} f(x) = -1$ and $\lim\limits_{x \to 0_+} f(x) = 1$

3. Show that f is discontinuous at $x = 0$

   Take $\varepsilon = 1$ Thus there exists $\delta > 0$ satisfying

   $$|x - 0| < \delta \implies |f(x) - f(0)| < \varepsilon$$

   Pick $x = -\dfrac{\delta}{2}$ Then $|x - 0| < \delta$. Since $x < 0$ we get that $f(x) = -1$ and $f(0) = 1$. So

   $$|f(x) - f(0)| = |-1 - 1| = 2 \not< 1$$

   A contradiction. Therefore f is discontinuous at $x = 0$

4. Show by definition that the function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \cos(x)$, is continuous on $\mathbb{R}$.

Fix $\varepsilon > 0$ and let $\xi \in \mathbb{R}$ be arbitrary. Choose $\delta = \varepsilon$. Then,

$$|f(x) - f(\xi)| = |\cos(x) - \cos(\xi)|$$

Using a stupid difference of cosines property

$$|\cos(x) - \cos(\xi)| = \left|2\sin\left(\frac{x+\xi}{2}\right)\sin\left(\frac{x-\xi}{2}\right)\right|$$

Since $\sin a \leq 1$ and $\sin a \leq a$ for all $a \in \mathbb{R}$

$$\leq \left|2\sin\left(\frac{x-\xi}{2}\right)\right| \leq \left|2\frac{x-\xi}{2}\right| = |x-\xi| < \delta = \varepsilon$$

Since ξ was arbitrary, $f(x)$ is continuous at every $\xi \in \mathbb{R}$, hence f is continuous on $\mathbb{R}$.

Homework 6

1. Show by definition that the function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \cos(x)$, is continuous on $\mathbb{R}$.

(This problem was done in Homework 5)

2. Show by definition that the function $f : [0, \infty) \to \mathbb{R}$ where $f(x) = \sqrt{x}$, is continuous on $\mathbb{R}$.

Suppose that $\xi > 0$ and fix $\varepsilon > 0$. Choose $\delta = \min\{\frac{\xi}{2}, \varepsilon\sqrt{\xi}\}$. Thus if $|x - \xi| < \delta$ we get that

$$\left|\sqrt{x} - \sqrt{\xi}\right| = \left|\sqrt{x} - \sqrt{\xi}\right| \cdot \left|\frac{\sqrt{x} + \sqrt{\xi}}{\sqrt{x} + \sqrt{\xi}}\right| = \left|\frac{x - \xi}{\sqrt{x} + \sqrt{\xi}}\right| = \frac{\delta}{\sqrt{x} + \sqrt{\xi}} < \frac{\delta}{\sqrt{\xi}} = \varepsilon$$

Thus f is continuous at every $\xi > 0$
Now suppose that $\xi = 0$. Since $f : [0, \infty) \to \mathbb{R}$ we can only approach 0 from the right. Thus we force $x > 0$. Choose $\delta = \varepsilon^2$ if $x - 0 < \delta$ we get that

$$\sqrt{x} - \sqrt{0} = \sqrt{x} < \sqrt{\delta} = \sqrt{\varepsilon^2} = \varepsilon$$

Thus f is continuous $\xi = 0$

3. Consider the function $f(x) : (0, \infty) \to \mathbb{R}$ where $f(x) = \frac{1}{x}$

1. Show by definition that f is continuous at every $\xi > 0$.

   Suppose that $\xi > 0$ and fix $\varepsilon > 0$. Choose $\delta = \min\{\frac{\xi}{2}, \frac{\xi^2}{2}\varepsilon\}$. Thus if $|x - \xi| < \delta$ (meaning $x < \xi/2$) we get that

   $$\left|\frac{1}{x} - \frac{1}{\xi}\right| = \left|\frac{\xi - x}{x\xi}\right| < \frac{\delta}{\left|\frac{\xi}{2}\xi\right|} = \delta\frac{2}{\xi^2} \leq \frac{2}{\xi^2}\left(\frac{\xi^2}{2}\varepsilon\right) = \varepsilon$$

2. Next suppose that the limit $\lim_{x \to 0} \frac{1}{x} = L$ exists. Then by definition for all $\varepsilon > 0$ there exists $\delta > 0$ such that

   $$x - 0 < \delta \implies \left|\frac{1}{x} - L\right| < \varepsilon$$

   Choose $\varepsilon = 1$ and $x = \min\{\frac{\delta}{2}, \frac{1}{|L|+1}\} > 0$. Thus if $x < \delta$ then $x < \frac{1}{1+|L|}$. Thus

   $$\left|\frac{1}{x} - L\right| \geq \frac{1}{x} - |L| \geq (|L| + 1) - |L| = 1 = \varepsilon$$

   Thus for $\varepsilon = 1$ there exists no $\delta > 0$ that works for any L.

4. Prove Theorem 2.4 via Theorem 2.2

2.2 Theorem: Let $f : I \to \mathbb{R}$ be a function. Let $\xi \in I$. Then f is continuous at ξ if and only iff the following statement is true: For all $\{x_n\}_{n=1}^{\infty}$ on I with $\lim_{n\to\infty} x_n = \xi, \{f(x_n)\}_{n=1}^{\infty}$ is convergent and $\lim_{n\to\infty} f(x_n) = f(\xi)$

2.4 Theorem: The product of continuous functions is continuous.

Since f, g are continuous at ξ then by 2.2 we get that for all $\{x_n\}$ on I with $\lim_{n\to\infty} x_n = \xi$, $\lim_{n\to\infty} f(x_n) = f(\xi)$, $\lim_{n\to\infty} g(x_n) = g(\xi)$,. By the product rule we get we get that $\{g \cdot f(x_n)\}$ is also a convergent sequence.

$$\lim_{n\to\infty} f(x_n) \cdot \lim_{n\to\infty} g(x_n) = \lim_{n\to\infty} [f(x_n)g(x_n)] = f \cdot g(\xi)$$

Thus by 2.2 $f \cdot g$ is continuous at ξ

Homework 7

1. Compute the following limit

$$\lim_{x\to 0}\frac{\sqrt{x^2+1}-1}{x}$$

Multiply the numerator and denominator by the conjugate

$$\frac{\sqrt{x^2+1}-1}{x}\cdot\frac{\sqrt{x^2+1}+1}{\sqrt{x^2+1}+1}=\frac{x^2+1-1}{x\sqrt{x^2+1}-x}=\frac{x}{\sqrt{x^2+1}+1}$$

Evaluating the equivalent limit we get

$$\lim_{x\to 0}\frac{x}{\sqrt{x^2+1}+1}=\frac{0}{\sqrt{0^2+1}+1}=0$$

2. Prove that if $f: I\to\mathbb{R}$ is continuous on I, then $|f|: I\to\mathbb{R}$ where

$$|f|(x):=|f(x)|\,\forall x\in I$$

is continuous on I. Then find a function $f(x)$ so that $|f(x)|$ is some continuous at some ξ while $f(x)$ is discontinuous at ξ

As a counter example of the converse, define $f(x)$ as

$$f(x)=\begin{cases}1 & \text{if } x\geq 0\\ -1 & \text{if } x<0\end{cases}$$

We proved that $f(x)$ is discontinuous at $\xi=0$ and we get that $|f(x)|\equiv 1$ which is a constant function, and thus continuous.
Suppose that f is continuous on I. Then for every $x\in I$ we get that for all $\varepsilon>0$ there exists $\delta>0$ such that

$$|x-\xi|<\delta \implies |f(x)-f(\xi)|<\varepsilon$$

If we choose the same $\delta>0$ and $\varepsilon>0$ we get that if $|x-\xi|<\delta$ we get that by inverse triangle inequality

$$||f(x)|-|f(\xi)||\leq|f(x)-f(\xi)|<\varepsilon$$

3. Show that the function $f(x)=\cos\left(\sqrt{x^2+\left(\frac{\pi}{2}\right)^2}\right)$ is continuous on $\mathbb{R}$, Then compute the following limit

$$\lim_{n\to\infty}\cos\left(\sqrt{\frac{1}{n^2}+\left(\frac{\pi}{2}\right)^2}\right)$$

(Thank GOD I can use composition of functions now)
Define $f_1, f_2 : \mathbb{R} \to [0, \infty)$ as $f_1(x) = x^2$ and $f_2(x) = (\frac{\pi}{2})^2$ which are both continuous. Define $f_3 : [0, \infty) \to [0, \infty)$ and $f_4 : \mathbb{R} \to [-1, 1]$ where $f_3(x) = \sqrt{x}$ and $f_4(x) = \cos x$. Recall that addition, multiplication, and composition of continuous functions is continuous. Thus we can write

$$f(x) = f_4 \circ f_3(f_1 + f_2)$$

Meaning f is continuous. As for the following limit recall that $\lim_{n\to\infty} \frac{1}{n^2} = 0$. Substituting we get

$$\lim_{n\to\infty} \cos\left(\sqrt{\frac{1}{n^2} + \left(\frac{\pi}{2}\right)^2}\right) = \cos\left(\sqrt{0 + \left(\frac{\pi}{2}\right)^2}\right) = \cos\frac{\pi}{2} = 0$$

4. Construct following three types of functions.

1. $f : I \to \mathbb{R}$ is continuous and I is bounded. But f is unbounded

   Let $I = (0, 1)$ and $f(x) = \frac{1}{x}$

2. $f : I \to \mathbb{R}$ where I is closed and bounded. But f is unbounded

   $I = [0, 1]$ and $f(x) = \frac{1}{x}$ (Note: it didn't say that f had to be continuous.)

3. $f : I \to \mathbb{R}$ is continuous and I is bounded. f has a minimum point but doesn't have a maximum point

   $I = (0, 1]$ and $f(x) = \frac{1}{x}$. F is monotonically decreasing and $f(1)$ exists, meaning $f(1)$ is the minimum. There is no maximum because for any $f(n)$ is the minimum we get that if $\xi < n$ then $f(\xi) > f(n)$

Homework 8

1. Let f be a continuous function defined on a closed, bounded interval $I = [a, b]$. Assume that f has a maximum M at some $\xi \in (a, b)$, then f cannot be one-to-one

Suppose $f(\xi) = M$ for some $a < \xi < b$ since (a, b) is an open interval. Where $M \geq f(x), \forall x \in (a, b)$. Choose u and v such that

$$a < u < \xi < v < b$$

Thus we get that $f(u), f(v) < M$. Next choose y such that

$$\max\{f(u), f(v)\} < y < M$$

Thus we have 2 variables $y_1 = y_2$ such that

$$f(u) < y < f(\xi) \quad \text{and} \quad f(v) < y < f(\xi)$$

If we assume f is one-to-one. Then buy the MVT and by continuity we get that there exists $s \in (u, \xi)$ and $t \in (\xi, v)$ such that $f(s) = y_1$ and $f(t) = y_2$. But that means $f(t) = f(s)$ where $t \neq s$. Thus f is not one-to-one.

2. Let f be a continuous function defined on a closed, bounded interval $I = [a, b]$. Assume f is one-to-one. Let m and M be the minimum and maximum of f respectively. Then my problem 1 we know that either $f(a) = m$ and $f(b) = M$ or $f(a) = M$ and $f(b) = m$. If $f(a) = m$ and $f(b) = M$, then show that f is strictly monotone increasing. If $f(a) = M$ and $f(b) = m$, then show that f is strictly monotone decreasing.

Suppose that $f(a) = m$ and $f(b) = M$. Let $x \in (a, b)$. On the interval $(x, b]$ we get that by question 1 and that by f being one-to-one, the maximum can only be obtained on b so $f(x) < f(b)$. Similarly on the interval $[a, x)$ the minimum can only be obtained on a. Thus for any $x \in (a, b)$ we get

$$f(a) < f(x) < f(b)$$

Assume for contradiction that for $x, y \in (a, b)$ where $a < x < y < b$ we get

$$f(a) < f(y) < f(x) < f(b)$$

By continuity of f we can choose c such that

$$f(a) < f(y) < c < f(x) < f(b)$$

Thus there exist u, v, w such that

1. $u \in (a, x) \implies f(u) = c$

2. $u \in (x, y) \implies f(v) = c$

3. $u \in (y, b) \implies f(w) = c$

Since u, v and w are in separate intervals and $f(u) = f(v) = f(w)$ we get that f is not one-to-one, a contradiction. Thus $x < y \implies f(x) < f(y)$ which is the definition of monotonically increasing.

To prove f monotone decreasing, let $g(x) := -f(x)$ thus $g(a) < g(b)$ and use the same proof for monotonically increasing. Thus it would suffice to show that if $g(x)$ is increasing, then $f(x) = -g(x)$ is decreasing. Recall for all x, y we get $g(x) < g(y)$. subtracting $g(x)$ and $g(y)$ from both sides we get that $-g(y) < -g(x)$ meaning $f(y) < f(x)$.

3. Consider $f : I \to \mathbb{R}$. Let J be an interval containing $f(I)$. Let $g : J \to \mathbb{R}$. Consider the composition of f and g, i.e. $g \circ f : I \to \mathbb{R}$. Show that:

1. If both f and g are strictly monotone increasing, then $g \circ f : I \to \mathbb{R}$ is strictly monotone increasing.

   By definition for all $a, b \in I$ then if $a < b$ then $f(a) < f(b)$. And for all $x, y \in J$ then if $x < y$ then $g(x) < g(y)$. Thus we get that

   $$x < y \implies f(x) < f(y) \implies g \circ f(x) < g \circ f(y)$$

   Thus $g \circ f$ is strictly monotone increasing

2. If both f and g are strictly monotone decreasing, then $g \circ f : I \to \mathbb{R}$ is strictly monotone increasing.

   By definition for all $a, b \in I$ then if $a < b$ then $f(b) < f(a)$. And for all $x, y \in J$ then if $x < y$ then $g(y) < g(x)$. Thus we get that

   $$x < y \implies f(y) < f(x) \implies g \circ f(x) < g \circ f(y)$$

   Thus $g \circ f$ is strictly monotone increasing

3. If one of f and g are strictly monotone increasing and the other is strictly monotone decreasing, then $g \circ f : I \to \mathbb{R}$ is strictly monotone increasing.

   Case 1: f is monotone increasing
   By definition for all $a, b \in I$ then if $a < b$ then $f(a) < f(b)$. And for all $x, y \in J$ then if $x < y$ then $g(y) < g(x)$. Thus we get that

   $$x < y \implies f(x) < f(y) \implies g \circ f(y) < g \circ f(x)$$

Case 2: g is monotone increasing
By definition for all $a, b \in I$ then if $a < b$ then $f(b) < f(a)$. And for all $x, y \in J$ then if $x < y$ then $g(x) < g(y)$. Thus we get that

$$x < y \implies f(y) < f(x) \implies g \circ f(y) < g \circ f(x)$$

Thus $g \circ f$ is strictly monotone decreasing

4. Let $f : \mathbb{R} \to \mathbb{R}$ to be a continuous, strictly monotone function. Let $f(\mathbb{R}) = \mathbb{R}$. Thus the inverse of f^{-1} is defined on the whole real line $\mathbb{R}$. Show that $f^{-1} : \mathbb{R} \to \mathbb{R}$ is continuous on $\mathbb{R}$.

Since f is monotone (WLOG say increasing) then we get that for all $x, y \in \mathbb{R}$ we get that

$$x < y \implies f(x) < f(y)$$

Next we claim that f^{-1} is also monotonically increasing. We observe that since f^{-1} is defined on all of $\mathbb{R}$, then there exists $f(x), f(y) \in \mathbb{R}$. And thus

$$f(x) < f(y) \implies f^{-1}(f(x)) < f^{-1}(f(y)) \implies x < y$$

This was proven by contradiction in a previous exercise. Thus f^{-1} is monotonic. Take some arbitrary interval $[a, b] \in \mathbb{R}$ where $f(I) = J$ and $J = [f(a), f(b)]$ (we showed that f must take closed sets to closed sets in a previous problem). Next take $x \in (f(a), f(b))$ and we get that $f^{-1}(x) \in (a, b)$. Hence there exits r such that $(f^{-1}(x) - r, f^{-1}(x) + r) \subset I$. Next fix $\varepsilon > 0$ such that $\varepsilon < r$. We get that

$$[f^{-1}(x) - \varepsilon, f^{-1}(x) + \varepsilon] \subset (f^{-1}(x) - r, f^{-1}(x) + r) \subset I$$

Since f^{-1} is onto, we get that there exist x_1 and x_2 such that

$$f^{-1}(x_0) - \varepsilon = f^{-1}(x_1) \quad f^{-1}(x_0) + \varepsilon = f^{-1}(x_2)$$

And since f^{-1} is also monotonic we get that $x_1 < x_0 < x_2$. Choose $\delta = \min\{|x_1 - x_0|, |x_2 - x_0|\}$
Thus if $|x_0 - \xi| < \delta$ then we get that

$$f^{-1}(x_1) < f^{-1}(\xi) < f^{-1}(x_2)$$

$$f^{-1}(x_0) - \varepsilon < f^{-1}(\xi) < f^{-1}(x_0) + \varepsilon$$

$$|f^{-1}(\xi) - f^{-1}(x_0)| < \varepsilon$$

Meaning that f^{-1} is continuous on any arbitrary interval $I \in \mathbb{R}$. Thus f^{-1} is continuous on all of $\mathbb{R}$.

Homework 9

1. We've proved in class the follow lemma: *between any two real numbers there is a rational number*. Use this fact to show by definition that $\mathbb{Q}' = \mathbb{R}$. In other words, every real number is a limit point of the set of rational numbers.

Recall by definition of a limit point that if $x \in \mathbb{R}$ is a limit point of some set $\mathbb{Q} \subset \mathbb{R}$ then for all $\varepsilon > 0$ there exists $x_n \in S \backslash x$ such that $|x_n - x| < \varepsilon$. Fix $\varepsilon > 0$. Let $x \in \mathbb{R}$ be arbitrary. Then $x + \varepsilon \in \mathbb{R}$. By the lemma there exists that $x_0 \in \mathbb{Q}$ such that $x < x_0 < x + \varepsilon$ Subtracting x throughout the inequality we get

$$0 < x_0 - x < \varepsilon$$

Meaning x is a limit point in $\mathbb{Q}$. Since x was arbitrary it means that any $x \in \mathbb{R}$ is a limit point.

2. Use the following strategy to show continuity of $f(x)$. Note for $a > 1$, we've showed that $f(x) = a^x$ is strictly monotone increasing on $\mathbb{R}$. Show that:

1. The sequence $\{a^{\frac{1}{n}}\}$ is monotone decreasing and bounded below by 1.

   If $a > 1$ then take integers $1 \leq n < m$. Then

   $$n < m \implies \frac{1}{n} > \frac{1}{m}$$

   Since the sequence $\frac{1}{n}$ is strictly decreasing, and $f(x) = a^x$ is strictly increasing, by problem 3 of homework 8 we get that $f\left(\frac{1}{x}\right) = a^{\frac{1}{x}}$ is strictly decreasing. Thus the sequence is also strictly decreasing. Thus for all $n \geq 1$ we get that

   $$0 < \frac{1}{n} \implies a^0 < a^{\frac{1}{n}}$$

   Where $a^0 = 1$. Thus the sequence is bounded below by 1.

2. 1 is the greatest lower bound of $\left\{a^{\frac{1}{n}}\right\}$. Thus

   $$\lim_{n\to\infty} a^{\frac{1}{n}} = 1$$

   Then show that for any $a > 0$, it holds that $\lim_{n\to\infty} a^{\frac{1}{n}} = \lim_{n\to\infty} a^{-\frac{1}{n}} = 1$.

Suppose that $b > 1$ is the greatest lower bound. Since $\{b^n\}$ must be strictly increasing then there is some N such that $b^N > a$. Since $x^{\frac{1}{N}}$ us increasing (as x increases) on $(0, \infty)$, then we get

$$b > a^{\frac{1}{N}}$$

Thus we get that b can't be a lower bound.
We already proved that the limit suffices for $a \geq 1$. If $0 < a < 1$ we let $b = \dfrac{1}{a}$ such that $b > 0$. Then we get

$$a^{-\frac{1}{n}} = \frac{1}{a}^{\frac{1}{n}} = b^{\frac{1}{n}} \to 1$$

and

$$a^{\frac{1}{n}} = \frac{1}{a^{-\frac{1}{n}}} = \frac{1}{b^n} \to \frac{1}{1} = 1$$

3. Use part 2 above and the monotonicity of $f(x) = a^x$ to show by definition that $f(x) = a^x$ is continuous at $x = 0$

   Our claim is that

   $$\lim_{x \to 0} a^x = a^0 = 1$$

   Fix $\varepsilon > 0$. And fix N such that we choose $\delta = \frac{1}{N}$. Since $a^{1/n} \to 1$ and $a^{-1/n} \to 1$ we get that for all $n > N$

   $$|a^{1/n} - 1| < \varepsilon \qquad |a^{-1/n} - 1| < \varepsilon$$

   thus we get that if $|x - 0| < \delta$ then we get that $-\dfrac{1}{N} < x < \dfrac{1}{N}$ and thus

   $$|a^x - 1| < \varepsilon$$

4. Use the formula $a^x a^y = a^{x+y}$ for all $x, y \in \mathbb{R}$ and part 3 to show that $f(x)$ is continuous on $\mathbb{R}$

   We want to show that $\lim\limits_{x \to \xi} a^x = a^\xi$ for all ξ in $\mathbb{R}$. Let $\xi \in \mathbb{R}$ be arbitrary. Fix $\varepsilon > 0$. Recall by part 3 that f is continuous at $\xi = 0$ and thus we can choose δ such that

   $$|t - 0| < \delta \implies |a^t - 1| < \frac{\varepsilon}{a^\xi}$$

   Thus if $|x - \xi| < \delta$ we get that by the above formula and part 3

   $$|a^x - a^\xi| = |a^\xi||a^{x-\xi} - a^{\xi-\xi}| = |a^\xi||a^{x-\xi} - 1|$$

   $$|a^\xi||a^{x-\xi} - 1| < a^\xi \frac{\varepsilon}{a^\xi} < \varepsilon$$

   Thus since ξ was arbitrary, we get that f is continuous on $\mathbb{R}$.

We've showed in class that for all $a > 0$ and all $x, y \in \mathbb{R}$ that

$$a^{x+y} = a^x a^y \quad \textbf{and} \quad (a^x)^y = a^{xy}$$

Fix $a > 0$ and $a \neq 1$ then we can define $f(x) = \log_a(x) : \mathbb{R}_+ \to \mathbb{R}$ to be the inverse function of $a^x : \mathbb{R} \to \mathbb{R}_+$. Use the formulas above and the definition of $f(x)$ to show that

$$\log_a(xy) = \log_a(x) + \log_a(y) \quad \textbf{and} \quad y \log_a(x) = \log_a(x^y)$$

Since we define $\log_a x$ as the inverse function of a^x we get that $\log_a(a^x) = x$ and $a^{\log_a x} = x$. Let $\log_a(x) + \log_a(y) = u$ for some real number u. Razing both sides over a and applying exponent rules we get.

$$a^{\log_a(x)+\log_a(y)} = a^u$$

$$a^{\log_a(x)} a^{\log_a(y)} = a^u$$

$$xy = a^u$$

Taking the $\log_a$ of both sides

$$\log_a(xy) = \log_a a^u = u$$

Meaning

$$\log_a(x) + \log_a(y) = \log_a(xy)$$

Next, letting $y \log_a(x) = v$ for some real number v. Razing both sides over a and applying exponent rules we get.

$$a^{y \log_a(x)} = a^v$$

$$\left(a^{\log_a(x)}\right)^y = a^v$$

$$x^y = a^v$$

Taking the $\log_a$ of both sides

$$\log_a x^y = \log_a a^v = v$$

Meaning

$$y \log_a(x) = \log_a(x^y)$$

For each $a \in \mathbb{R}$ and $x > 0$, we have defined what is x^a. Fix $a \in \mathbb{R}$ and we consider x^a as a function in $x \in \mathbb{R}_+$, i.e.

$$f_a(x) = x^a : \mathbb{R}_+ \to \mathbb{R}$$

1. Use problem 3 to show that $x^a = e^{a \ln x}$ for all $x > 0$

   Fix $x > 0$ Recall that e^a and $\ln a$ are inverse functions. Meaning $a = e^{\ln a}$. And we also know that $\ln x^a = a \ln x$ for $x > 0$. Thus we get that

   $$x^a = e^{\ln x^a} = e^{a \ln x}$$

2. Use the above equality to show that $f_a(x) = x^a$ is continuous on $\mathbb{R}_+$

   We let $\xi \in \mathbb{R}_+$ (meaning $\ln \xi$ is defined). We want to show that for some $a \in \mathbb{R}$

   $$\lim_{x \to \xi} x^a = \xi^a$$

   Fix $a \in \mathbb{R}$ and $\varepsilon > 0$. Recall from part 3 we can choose δ such that

   $$\left| a \ln \frac{\xi}{x} - 0 \right| < \delta \implies \left| e^{a \ln \frac{\xi}{x}} - 1 \right| < \frac{\varepsilon}{|e^{a \ln \xi}|}$$

   Choose $\delta = \min\{\frac{\xi}{2}, \delta \text{ from part 3}\}$. Thus if $|x - \xi| < \delta$, we get that

   $$|x^a - \xi^a| = \left| e^{a \ln x} - e^{a \ln \xi} \right| = \left| e^{a \ln \xi} \right| \left| e^{a \ln x - a \ln \xi} - 1 \right|$$

   $$\left| e^{a \ln \xi} \right| \left| e^{a \ln \frac{x}{\xi}} - 1 \right| < \left| e^{a \ln \xi} \right| \frac{\varepsilon}{|e^{a \ln \xi}|} = \varepsilon$$

   Thus since ξ was arbitrary, we get that f is continuous on $\mathbb{R}_+$.

3. Again use the quality from part 1 to show that: if $a > 0$, then $f_a(x) = a^x$ is strictly monotone increasing; if $a < 0$, then $f_a(x) = x^a$ is strictly monotone decreasing.

   Suppose that $a > 0$. Recall that $\ln x$ is strictly monotone increasing and e^{ax} is also strictly monotone increasing for $a > 0$. Recall that composition of two strictly monotonically increasing functions is also strictly monotonically increasing. Thus we get that

   $$e^{a \ln x} = x^a$$

   is increasing
   Next, suppose $a < 0$. Then let $-b := a$ such that $b > 0$. Recall from the first part of the proof that if $x > y$ then $x^a > y^a$. Thus we get that if $x > y$

   $$x > y \implies \frac{1}{x} < \frac{1}{y} \implies \frac{1}{x^b} < \frac{1}{y^b}$$

   Recall that $\frac{1}{x^b} = x^a$ we get

   $$x^a < y^a$$

   Thus x^a is monotonically decreasing for x^a

4. By part 3, we know that $f_a(x)$ has an inverse for any $a \neq 0$. Compute it's inverse.

   Define a function $g_a(x) = x^{\frac{1}{a}}$. Composing the functions we get

   $$f_a \circ g_a(x) = \left(x^{\frac{1}{a}}\right)^a = x^{\frac{a}{a}} = x$$

   Thus by definition of the inverse g_a is the inverse of f

Closing remarks

I've taken all of the quizzes on my own time on physical paper to actually test what I know and I'm confident that I have all of these topics down. While some of my proofs leave some things to be desired (I overcomplicated simple proofs and oversimplify more complicated proofs), I just needed something to tease the fundamentals. Most of where I initially learned this stuff was from Conway's complex analysis book and YouTube videos so I had some experience in higher math.